Ethan PUYAUBREAU

C++ / HPC Software Engineer
DevOps • EDF R&D • Oak Ridge (USA)
M.Eng.-equivalent degree, 09/2026
Available from 01/2027

CONTACT

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TECHNICAL SKILLS

Specialties

HPC · Performance portability (Kokkos)
GPU energy & performance measurement
Scientific & research software engineering
Nuclear simulation software · DevOps

HPC & Scientific Computing

C++17 · Kokkos · Kokkos Tools · CMake
NVML · ROCm-SMI · Variorum · Slurm
Perfetto · Basics: CUDA, MPI, OpenMP

Infrastructure & Platform

Kubernetes (K3s) · Docker · Ceph
Proxmox VE · VyOS · Ansible · Traefik
Linux / Debian · Grafana · Prometheus

CI/CD & DevOps

GitLab CI/CD · Jenkins · GitHub Actions
Debian packaging · Git

Programming Languages

C++ · Python · TypeScript · Java · Bash
Rust (working knowledge) · Learning: Go

Data, Backend & Docs

PostgreSQL · MySQL · JSON/XML/XSD
LaTeX · Typst

LANGUAGES

French — Native
English — C1 — TOEIC 970/990 (10/2025)
Spanish — B1

BACKGROUND

Full legal name

Ethan Nicolas PUYAUBREAU

Citizenship

French

U.S. immigration status

In the U.S. on a B-1/B-2 visa, 10/2026 – 01/2027
Prior J-1 Short-Term Scholar (ORAU/ORNL), 06/2025 – 08/2025, not subject to INA 212(e)
Requires H-1B or J-1 sponsorship

Residence

France, except 06/2025 – 08/2025:
Knoxville, TN, USA (ORNL research stay)

Affiliations

EDF R&D: 04/2022 – 06/2022, 09/2023 – 09/2026
Université Paris-Saclay: 09/2019 – 09/2026
ORNL: 06/2025 – 08/2025
Kokkos Tools contributor: 06/2025 – Present

INTERESTS

Flight simulation (DCS World, KSP, VR)
3D & VFX · Music · Self-hosting

PROFILE

C++ / HPC software engineer trained through a three-year apprenticeship at EDF R&D (Électricité de France), on the platform that simulates EDF's nuclear reactor cores, with a research stay at Oak Ridge National Laboratory (USA). Dual background: high-performance computing (profiling, GPU, Kokkos) and infrastructure/DevOps (Kubernetes, CI/CD, Debian packaging). **Available full-time from 01/2027**.

WORK EXPERIENCE

Career Break After Graduation

Open source & personal projects — 10/2026 – Present

Contributions to Kokkos Tools (one PR merged and one opened in 10/2026), travel, job search.

Apprentice Engineer — Software Architecture & Scientific Computing

EDF R&D • EDF Lab Paris-Saclay, Palaiseau, France — 09/2023 – 09/2026

- Developed a prototype for nuclear reactor core computation following a more modular, component-oriented paradigm: results rigorously identical to the reference, **bit-for-bit**, up to **12% faster** in the computational core and **38% lower peak memory** (neutronic solvers, C++).
- Built two performance diagnostic tools for the simulation platform (**500k+ lines of code**, **30+ engineers**): memory profiling and compute-time measurement, used to benchmark the prototype; the memory tool pinpointed a blow-up the team had chased for days.
- Automated the test and delivery chain (Jenkins, GitLab CI/CD): test runs on clusters, result validation (XML/XSD) and storage (PostgreSQL), Debian packaging for Scibian, EDF's Debian derivative.
- Teamwork: code reviews given and received, leading technical meetings, user support; **5 internal technical notes**, **3 apprenticeship reports** with annual defenses.

C++17 · Python · pybind11 · CMake · PostgreSQL · GitLab CI/CD · Jenkins · Linux Scibian

Graduate Research Fellow — Graduate Research at ORNL (GRO)

Oak Ridge National Laboratory • CSED, Oak Ridge, TN, USA — 06/2025 – 08/2025

- Research stay during the apprenticeship. Built two tools measuring GPU energy per Kokkos kernel: a multi-vendor NVIDIA/AMD connector built on Variorum (NVML, ROCm-SMI) and a higher-precision NVIDIA-only tool; the ROCm-SMI version ran on **Frontier**, the world's first exascale supercomputer.
- Case study on a GPU clustering algorithm (ArborX DBSCAN, one H100, medians of 64 runs): the dense variant took **19% less time and 25% less energy**; a time-only profile underestimates the energy gain.
- Open-source contributions, ongoing since the stay: **4 merged pull requests** to Kokkos Tools and LAMMPS (11 opened); preliminary exploration of multi-node simulations (LAMMPS, MPI).
- Poster *Understanding GPU Energy Dynamics in HPC Applications* (first author, with D. Arndt, J. Bludau, D. Lebrun-Grandié), **presented twice** (Smoky Mountains Conference 2025, ORNL internal session); results reported back to EDF Lab department group leaders; invited by ORNL to present at SC25, declined.

C++17 · Kokkos · NVML · ROCm-SMI · Variorum · Perfetto · MPI · Slurm

Software Engineering Intern

EDF R&D • EDF Lab Paris-Saclay, Palaiseau, France — 04/2022 – 06/2022

Fault trees and bow-tie models in Python for nuclear power plant risk analysis (prototype).

EDUCATION

Diplôme d'Ingénieur (M.Eng. equivalent) — Computer Science & Applied Mathematics

Polytech Paris-Saclay • Université Paris-Saclay, France — 09/2023 – 09/2026

Master's-level degree, CTI-accredited, apprenticeship track. Scientific computing · HPC · Parallel programming · Software architecture · Networks & security · Machine learning. Final apprenticeship report and defense (09/2026) on high-performance computing and software architecture for a nuclear reactor core simulation platform (EDF, confidential); advisors: Joël Gay (Polytech), Ivan Dutka-Malen (EDF).

Bachelor of Science (Licence) — Science and Technology

Université Paris-Saclay • Institut Villebon – Georges Charpak, Orsay, France — 09/2019 – 06/2023

180 ECTS (enrolled at Université Paris-Sud, now Paris-Saclay). Mathematics (numerical analysis), physics (electromagnetism, fluid mechanics), computer science (graph theory, data analysis).

PROJECTS

Personal Multi-Server Infrastructure

Personal project • self-hosted — 06/2020 – Present

Designed and operated end to end, solo: Proxmox VE virtualization, Kubernetes (K3s) orchestration, Ceph distributed storage, segmented network (VyOS, VLANs), monitoring and automated deployment (GitLab CI/CD, Ansible). **Around twenty services running continuously** for about 60 regular users.

Proxmox VE · Kubernetes · Ceph · Docker · VyOS · Ansible · GitLab CI/CD

Vireli — Carbon Footprint Web App (team lead)

Polytech Paris-Saclay • industry partnership with Viveris — 09/2025 – 05/2026

Led a team of six (554 project hours) and owned the architecture, backend (Hono, Bun, Prisma) and deployment (Gitea CI, Coolify, PostgreSQL, Redis). **18/18 functional requirements delivered** and validated.

TypeScript · PostgreSQL · Docker · CI/CD · Team lead

Large-Scale Online Event Organization

Volunteer organizer — 01/2021 – Present

Multiplayer events reaching **150 to 180+ participants**, recognized by the specialist press. Coordination of a staff of four to five, live technical monitoring (Grafana, Prometheus), systematic debriefs.